

Assignment No. 2

EECS 690

Introduction to Machine Learning

Due: 11:59 PM, Friday, September 18, 2020

Submit deliverables in a single zip file to BlackBoard

Name of the zip file: FirstnameLastname_Assignment2

Name of the Assignment folder within the zip file: FirstnameLastname_Assignment2

Deliverables:

1. Copy of Rubric2.docx with your name and ID filled out (do not submit a PDF)
2. Python source code for CompareMLModels
3. Screen print showing the successful execution of CompareMLModels
4. Answers to the following questions:
 - a. Based on accuracy which model is the best one?
 - b. For each of the 6 other models, explain why you think it does not perform as well as the best one.

Assignment:

- For this assignment, we are going to compare how well different ML classifiers classify the iris dataset.
- Write a Python program called CompareMLModels that does the following:
 - Uses 2-fold cross-validation to produce a test set of 150 samples of the iris data set with the following ML models:
 - Naïve Baysian (NBClassifier)
 - Linear regression (LinearRegression)
 - Polynomial of degree 2 regression (LinearRegression)
 - Polynomial of degree 3 regression (LinearRegression)
 - kNN (KNeighborsClassifier)
 - LDA (LinearDiscriminantAnalysis)
 - QDA (QuadraticDiscriminantAnalysis)
 - Remember 2-fold cross-validation involves:
 - Dividing the data set into 2 folds
 - Training the model with fold 1
 - Testing the model with fold 2
 - Training the model with fold 2
 - Testing the model with fold 1
 - Concatenating the test results from the 2 folds to get a test set of 150 samples.
 - For each of the 7 models the program should display (with a label before each model's display indicating which model the results are for):
 - Confusion matrix
 - Accuracy metric
 - If the values in your confusion matrices do not add up to 150, then you did something wrong.

Remember:

- Your Programming Assignments are individual-effort.
- You can brainstorm with other students and help them work through problems in their programs, but everyone should have their own unique assignment programs.